Model examples

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Introduction

This vignette demonstrates a variety of epidemiological models built with `algebraicodin`, showcasing Petri net, stock-flow, and compositional (AlgebraicDynamics-style) approaches. Each model is generated as odin2 code, compiled, and simulated.

```
library(algebraicodin)
library(catlab)
library(ggplot2)
library(odin2)
library(dust2)
```

1. SIRS model (waning immunity)

The SIRS model extends SIR with waning immunity: recovered individuals return to the susceptible class.

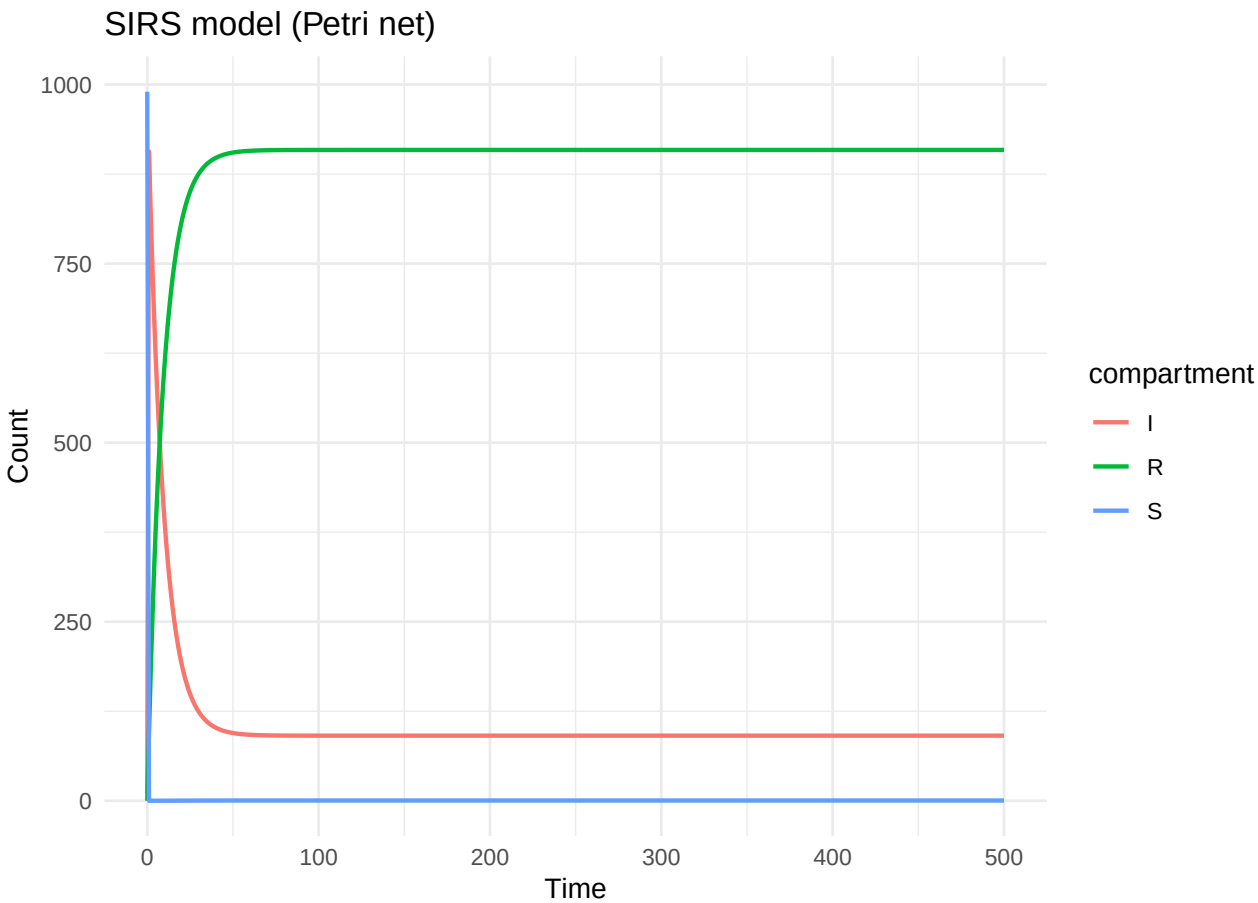
Petri net approach

```
sirs_pn <- labelled_petri_net(
  c("S", "I", "R"),
  infection = c("S", "I") %=>% c("I", "I"),
  recovery = "I" %=>% "R",
  waning = "R" %=>% "S"
)

code <- pn_to_odin(sirs_pn)
cat(code)
#> ## Auto-generated by algebraicodin
#>
#> ## Parameters
#> infection <- parameter()
#> recovery <- parameter()
#> waning <- parameter()
#>
#> ## Initial conditions
#> S0 <- parameter()
#> I0 <- parameter()
#> R0 <- parameter()
#>
#> initial(S) <- S0
#> initial(I) <- I0
#> initial(R) <- R0
#>
```

```
#> ## Transition rates
#> rate_infection <- infection * S * I
#> rate_recovery <- recovery * I
#> rate_waning <- waning * R
#>
#> ## Derivatives
#> deriv(S) <- -rate_infection + rate_waning
#> deriv(I) <- rate_infection - rate_recovery
#> deriv(R) <- rate_recovery - rate_waning
```

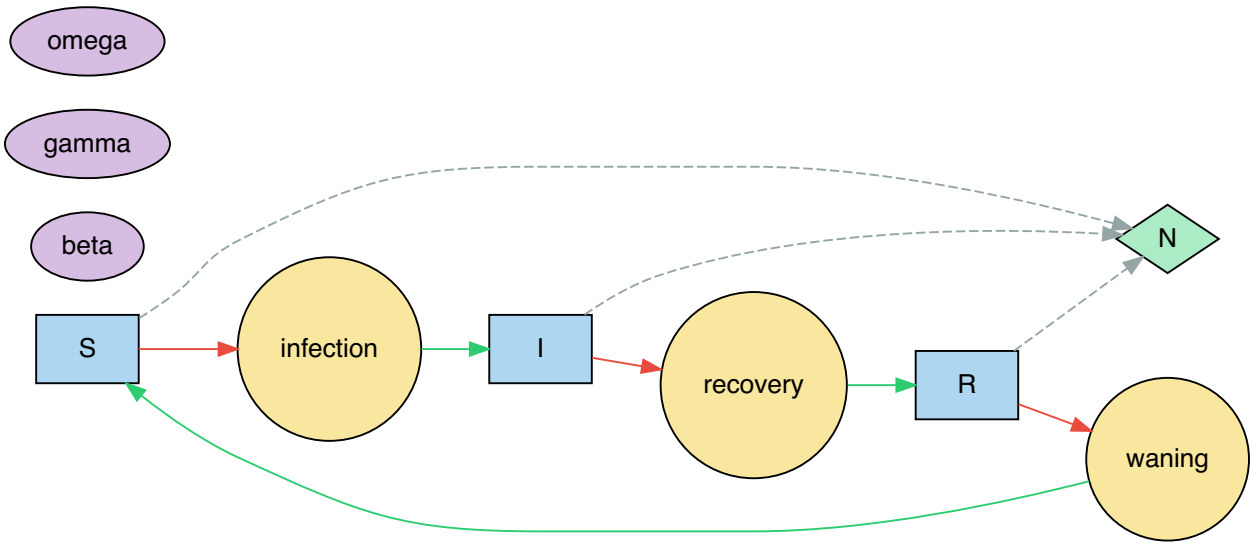
```
gen <- pn_to_odin_system(sirs_pn)
sys <- dust_system_create(gen(), list(
  S0 = 990, I0 = 10, R0 = 0,
  infection = 0.3, recovery = 0.1, waning = 0.01
))
#> -- R CMD INSTALL ------------------------------------------------------------------------------------
#>   -  installing *source* package ‘odin.system4d017138’ ... (369ms)
#>    ** this is package ‘odin.system4d017138’ version ‘0.0.1’
#>      ** using staged installation
#>      ** libs
#>      using C++ compiler: ‘Homebrew clang version 21.1.5’
#>      using SDK: ‘MacOSX15.5.sdk’
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      In file included from dust.cpp:85:
#>    In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/random
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>      /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/rand
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/include/dust2/r/contin
#>       34 |   auto seed = monty::random::r::as_rng_seed<rng_state_type>(r_seed);
#>          |                                     ^
#>    dust.cpp:89:20: note: in instantiation of function template specialization 'dust2::r::dust2_conti
#>     89 |   return d        89 |   return dust2::r::dust2_continuous_alloc<odin_system>(r_pars, r_tim
#>          |                    ^
#>      2 warnings generated.
#>      clang++ -arch arm64 -std=gnu++17 -dynamiclib -Wl,-headerpad_max_install_names -undefined dynami
#>      installing to /private/var/folders/yh/30rj513j6mn1n7x556c2v4w80000gn/T/RtmpLWyqXP/devtools_inst
#>      ** checking absolute paths in shared objects and dynamic libraries
#>   -  DONE (odin.system4d017138) (432ms)
#>
dust_system_set_state_initial(sys)
times <- seq(0, 500, by = 1)
res <- dust_system_simulate(sys, times)
state <- dust_unpack_state(sys, res)

df <- data.frame(
  time = rep(times, 3),
  value = c(state$S, state$I, state$R),
  compartment = rep(c("S", "I", "R"), each = length(times))
)
```

```r
ggplot(df, aes(x = time, y = value, colour = compartment)) +
  geom_line(linewidth = 0.8) +
  labs(title = "SIRS model (Petri net)", x = "Time", y = "Count") +
  theme_minimal()
```



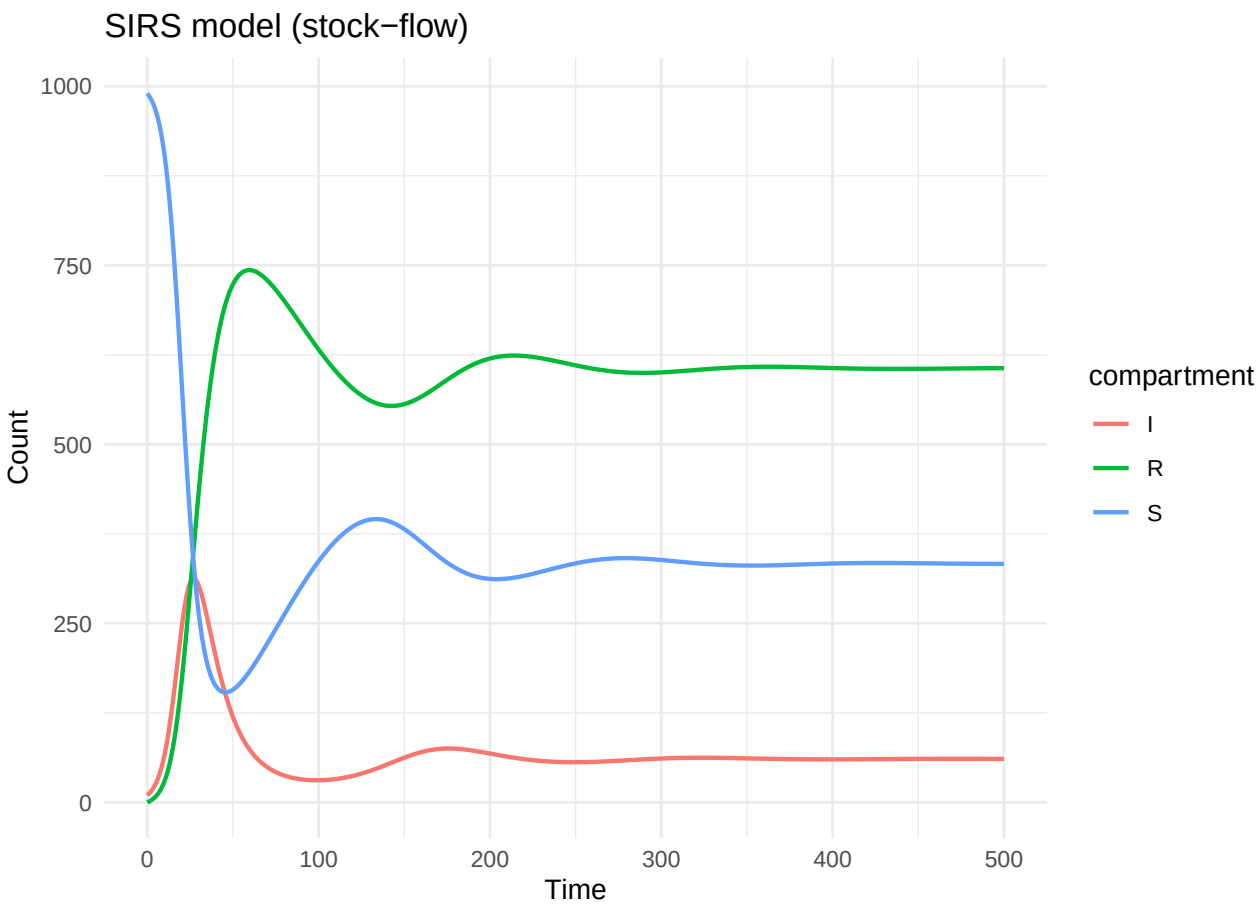
Stock-flow approach

```r
sirs_sf <- stock_and_flow(
  stocks = c("S", "I", "R"),
  flows = list(
    infection = flow(from = "S", to = "I", rate = beta * S * I / N),
    recovery = flow(from = "I", to = "R", rate = gamma * I),
    waning = flow(from = "R", to = "S", rate = omega * R)
  ),
  sums = list(N = c("S", "I", "R")),
  params = c("beta", "gamma", "omega")
)

plot_stock_flow(sirs_sf)
#> Google Chrome was not found. Try setting the `CHROMOTE_CHROME` environment variable to the executabl
```




```
gen <- sf_to_odin_system(sirs_sf, type = "ode")
sys <- dust_system_create(gen(), list(
  beta = 0.3, gamma = 0.1, omega = 0.01,
  S0 = 990, I0 = 10, R0 = 0
))
#> -- R CMD INSTALL ------------------------------------------------------------------------------------
#>   -  installing *source* package ‘odin.systemb6dd3362’ ...
#>    ** this is package ‘odin.systemb6dd3362’ version ‘0.0.1’
#>      ** using staged installation
#>      ** libs
#>      using C++ compiler: ‘Homebrew clang version 21.1.5’
#>      using SDK: ‘MacOSX15.5.sdk’
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      In file included from dust.cpp:86:
#>    In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/random
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>      /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/rand
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/include/dust2/r/contin
#>       34 |   auto seed = monty::random::r::as_rng_seed<rng_state_type>(r_seed);
#>          |                                            ^
#>    dust.cpp:90:20: note: in instantiation of function template specialization 'dust2::r::dust2_conti
#>     90 |   return dust2::r::dust2_continuous_alloc<odin_system>(r_pars, r_time, r_time_control, r_n_
#>          |                    ^
#>      2 warnings generated.
#>      clang++ -arch arm64 -std=gnu++17 -dynamiclib -Wl,-headerpad_max_install_names -undefined dynami
#>      installing to /private/var/folders/yh/30rj513j6mn1n7x556c2v4w80000gn/T/RtmpLWyqXP/devtools_inst
#>      ** checking absolute paths in shared objects and dynamic libraries
#>   -  DONE (odin.systemb6dd3362) (414ms)
#>
dust_system_set_state_initial(sys)
res <- dust_system_simulate(sys, times)
```

```r
state <- dust_unpack_state(sys, res)

df <- data.frame(
  time = rep(times, 3),
  value = c(state$S, state$I, state$R),
  compartment = rep(c("S", "I", "R"), each = length(times))
)

ggplot(df, aes(x = time, y = value, colour = compartment)) +
  geom_line(linewidth = 0.8) +
  labs(title = "SIRS model (stock-flow)", x = "Time", y = "Count") +
  theme_minimal()
```

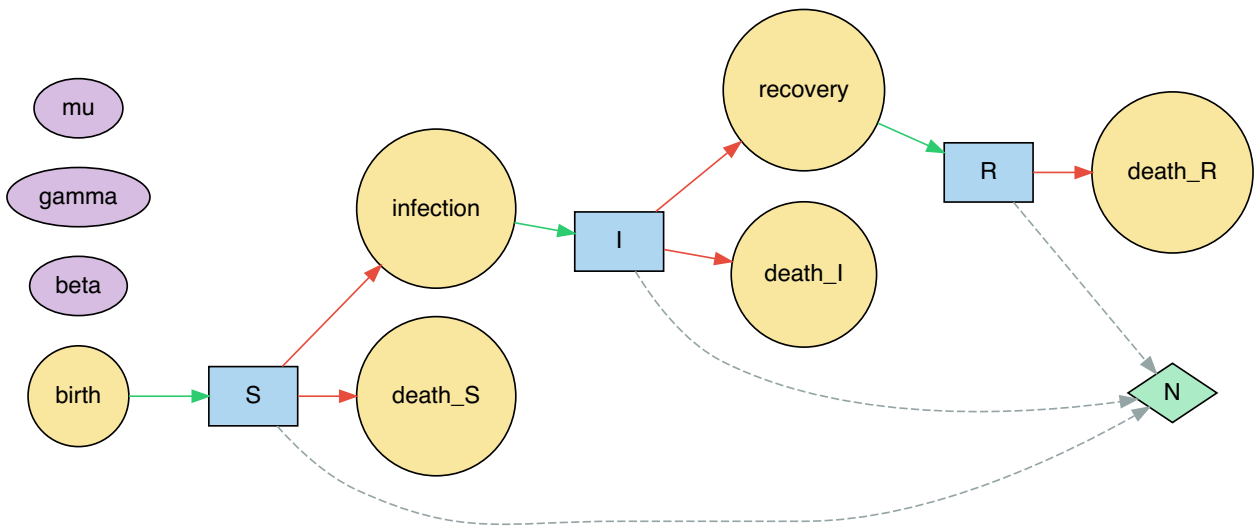


2. SIR with demographics (births and deaths)

An open-population SIR model where the total population is maintained by births and natural deaths.

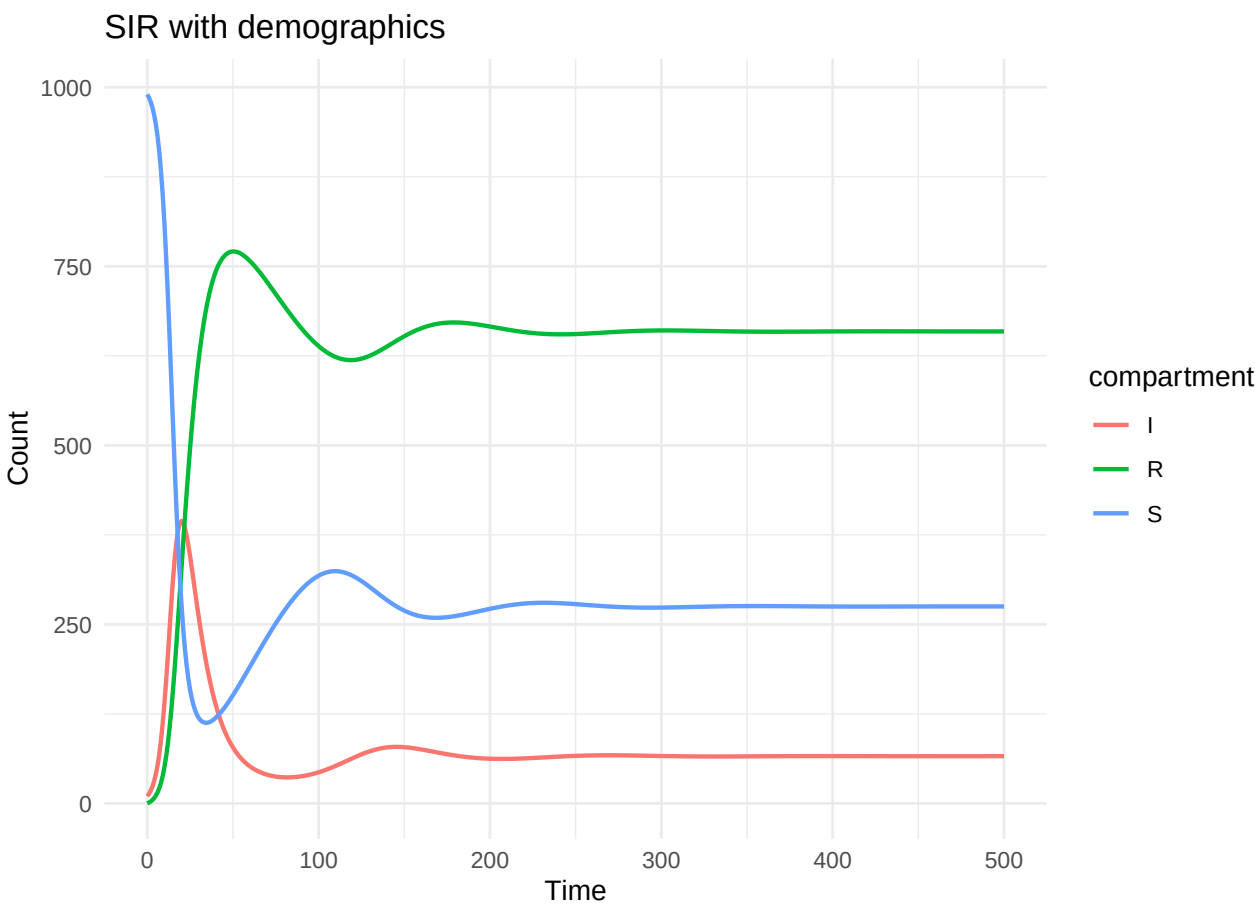
```r
sir_demog <- stock_and_flow(
  stocks = c("S", "I", "R"),
  flows = list(
    birth = flow(from = NULL, to = "S", rate = mu * N),
    infection = flow(from = "S", to = "I", rate = beta * S * I / N),
    recovery = flow(from = "I", to = "R", rate = gamma * I),
    death_S = flow(from = "S", to = NULL, rate = mu * S),
    death_I = flow(from = "I", to = NULL, rate = mu * I),
```

```r
    death_R = flow(from = "R", to = NULL, rate = mu * R)
  ),
  sums = list(N = c("S", "I", "R")),
  params = c("beta", "gamma", "mu")
)

plot_stock_flow(sir_demog)
```




```r
gen <- sf_to_odin_system(sir_demog, type = "ode")
sys <- dust_system_create(gen(), list(
  beta = 0.4, gamma = 0.1, mu = 0.01,
  S0 = 990, I0 = 10, R0 = 0
))
#> -- R CMD INSTALL ------------------------------------------------------------------------------------
#>   -  installing *source* package ‘odin.systemd434ad23’ ... (336ms)
#>    ** this is package ‘odin.systemd434ad23’ version ‘0.0.1’
#>      ** using staged installation
#>      ** libs
#>      using C++ compiler: ‘Homebrew clang version 21.1.5’
#>      using SDK: ‘MacOSX15.5.sdk’
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      In file included from dust.cpp:89:
#>    In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/random
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>      /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/rand
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/include/dust2/r/contin
#>       34 |   auto seed = monty::random::r::as_rng_seed<rng_state_type>(r_seed);
#>          |                                           ^
#>    dust.cpp:93:20: note: in instantiation of function template specialization 'dust2::r::dust2_conti
#>       93 |   return dust2::r::dust2_continuous_alloc<odin_system>(r_pars, r_time, r_time_control, r_
#>          |                     ^
#>      2 warnings generated.
```

```
#>      clang++ -arch arm64 -std=gnu++17 -dynamiclib -Wl,-headerpad_max_install_names -undefined dynami
#>      installing to /private/var/folders/yh/30rj513j6mn1n7x556c2v4w80000gn/T/RtmpLWyqXP/devtools_inst
#>      ** checking absolute paths in shared objects and dynamic libraries
#>   -  DONE (odin.systemd434ad23) (461ms)
#>
dust_system_set_state_initial(sys)
times <- seq(0, 500, by = 1)
res <- dust_system_simulate(sys, times)
state <- dust_unpack_state(sys, res)

df <- data.frame(
  time = rep(times, 3),
  value = c(state$S, state$I, state$R),
  compartment = rep(c("S", "I", "R"), each = length(times))
)

ggplot(df, aes(x = time, y = value, colour = compartment)) +
  geom_line(linewidth = 0.8) +
  labs(title = "SIR with demographics", x = "Time", y = "Count") +
  theme_minimal()
```



3. SEIR model

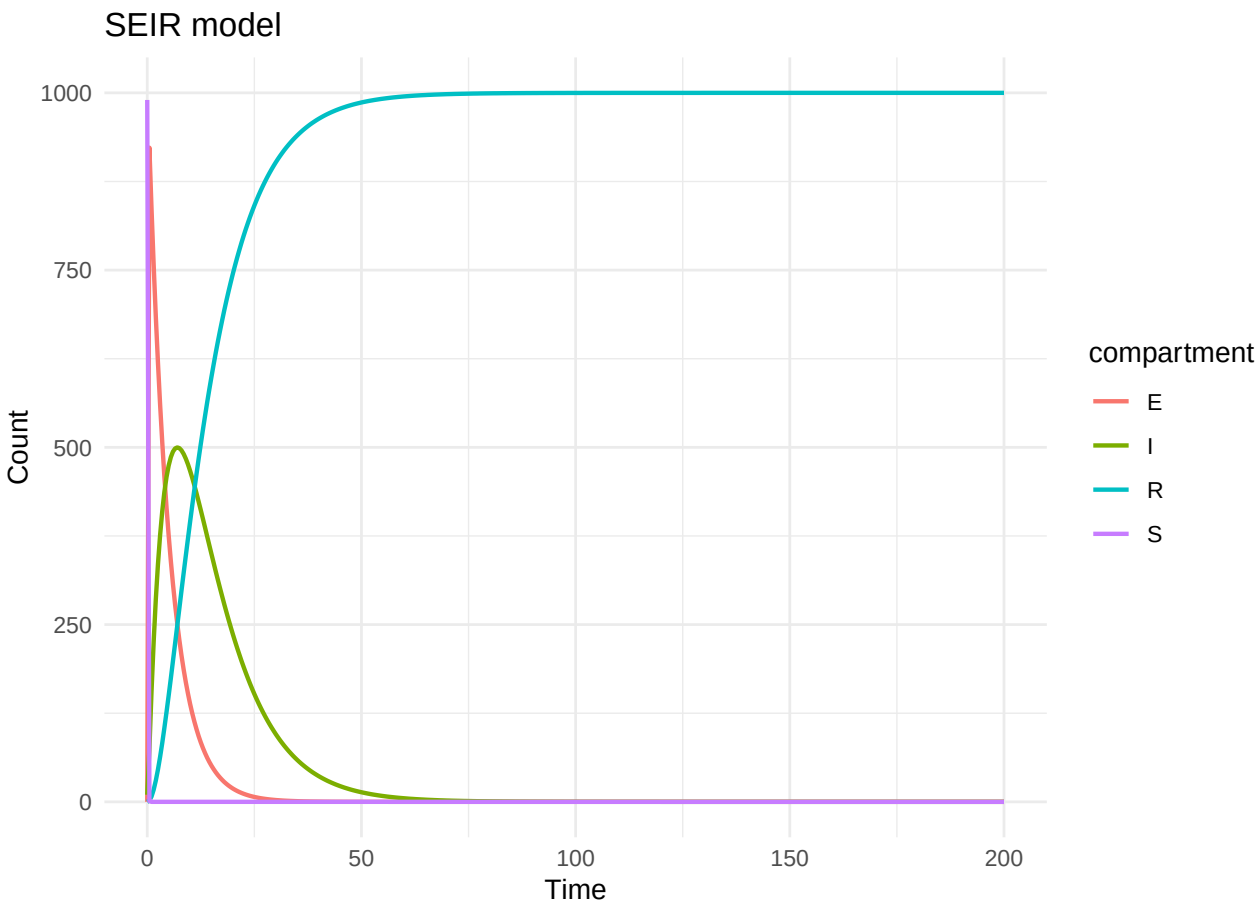
The SEIR model includes an exposed (latent) class between susceptible and infectious.

```r
seir_pn <- labelled_petri_net(
  c("S", "E", "I", "R"),
  exposure = c("S", "I") %=>% c("E", "I"),
  infection = "E" %=>% "I",
  recovery = "I" %=>% "R"
)

code <- pn_to_odin(seir_pn)
cat(code)
#> ## Auto-generated by algebraicodin
#>
#> ## Parameters
#> exposure <- parameter()
#> infection <- parameter()
#> recovery <- parameter()
#>
#> ## Initial conditions
#> S0 <- parameter()
#> E0 <- parameter()
#> I0 <- parameter()
#> R0 <- parameter()
#>
#> initial(S) <- S0
#> initial(E) <- E0
#> initial(I) <- I0
#> initial(R) <- R0
#>
#> ## Transition rates
#> rate_exposure <- exposure * S * I
#> rate_infection <- infection * E
#> rate_recovery <- recovery * I
#>
#> ## Derivatives
#> deriv(S) <- -rate_exposure
#> deriv(E) <- rate_exposure - rate_infection
#> deriv(I) <- rate_infection - rate_recovery
#> deriv(R) <- rate_recovery
```

```r
gen <- pn_to_odin_system(seir_pn)
sys <- dust_system_create(gen(), list(
  S0 = 990, E0 = 0, I0 = 10, R0 = 0,
  exposure = 0.4, infection = 0.2, recovery = 0.1
))
#> -- R CMD INSTALL ------------------------------------------------------------------------------
#>   -  installing *source* package ‘odin.systemca2473fd’ ...
#>    ** this is package ‘odin.systemca2473fd’ version ‘0.0.1’
#>      ** using staged installation
#>      ** libs
#>      using C++ compiler: ‘Homebrew clang version 21.1.5’
#>      using SDK: ‘MacOSX15.5.sdk’
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      In file included from dust.cpp:92:
#>    In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/
```

```
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/random
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>      /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/rand
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/include/dust2/r/contin
#>       34 |   auto seed = monty::random::r::as_rng_seed<rng_state_type>(r_seed);
#>          |                                         ^
#>    dust.cpp:96:20: note: in instantiation of function template specialization 'dust2::r::dust2_conti
#>       96 |   return dust2::r::dust2_continuous_alloc<odin_system>(r_pars, r_time, r_time_control, r_
#>          |                      ^
#>      2 warnings generated.
#>      clang++ -arch arm64 -std=gnu++17 -dynamiclib -Wl,-headerpad_max_install_names -undefined dynami
#>      installing to /private/var/folders/yh/30rj513j6mn1n7x556c2v4w80000gn/T/RtmpLWyqXP/devtools_inst
#>      ** checking absolute paths in shared objects and dynamic libraries
#>   -  DONE (odin.systemca2473fd) (422ms)
#>
```

```r
dust_system_set_state_initial(sys)
times <- seq(0, 200, by = 0.5)
res <- dust_system_simulate(sys, times)
state <- dust_unpack_state(sys, res)

df <- data.frame(
  time = rep(times, 4),
  value = c(state$S, state$E, state$I, state$R),
  compartment = rep(c("S", "E", "I", "R"), each = length(times))
)

ggplot(df, aes(x = time, y = value, colour = compartment)) +
  geom_line(linewidth = 0.8) +
  labs(title = "SEIR model", x = "Time", y = "Count") +
  theme_minimal()
```



4. Compositional SIR using resource sharers

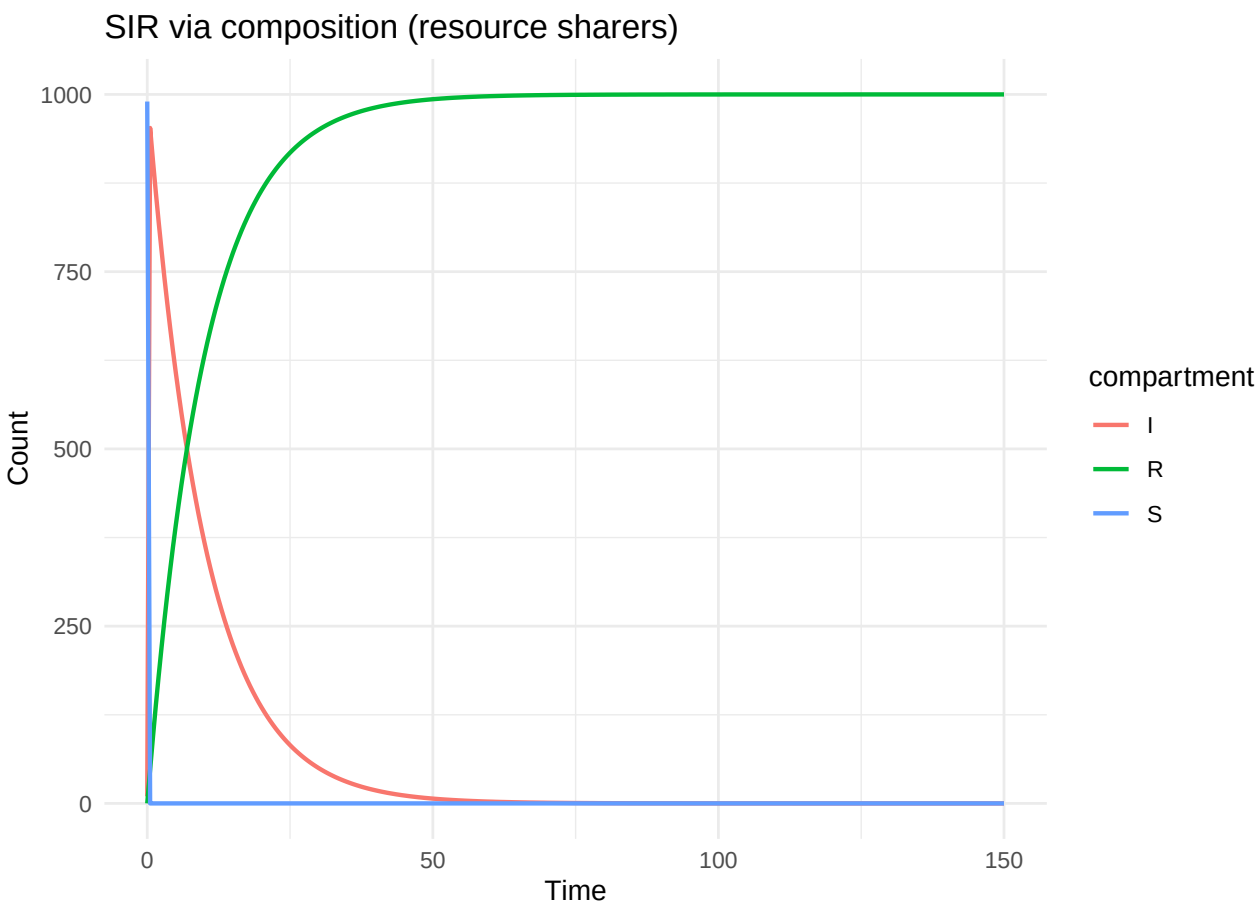
The same SIR model can be built by composing independent sub-models:

```
infection_rs <- continuous_resource_sharer(
  state_names = c("S", "I"),
  dynamics_expr = list(S = quote(-beta * S * I), I = quote(beta * S * I)),
  params = c("beta")
)

recovery_rs <- continuous_resource_sharer(
  state_names = c("I", "R"),
  dynamics_expr = list(I = quote(-gamma * I), R = quote(gamma * I)),
  params = c("gamma")
)

sir_uwd <- uwd(
  outer = c("s", "i", "r"),
  infection = c("s", "i"),
  recovery = c("i", "r")
)

sir_composed <- oapply_dynam(sir_uwd, list(
  infection = infection_rs,
  recovery = recovery_rs
```

```r
))

code <- rs_to_odin(sir_composed, type = "ode",
  initial = c(S = 990, I = 10, R = 0))
cat(code)
#> ## Auto-generated by algebraicodin (ResourceSharer)
#>
#> ## Parameters
#> beta <- parameter()
#> gamma <- parameter()
#>
#> ## Initial conditions
#> S0 <- parameter(990)
#> I0 <- parameter(10)
#> R0 <- parameter(0)
#>
#> initial(S) <- S0
#> initial(I) <- I0
#> initial(R) <- R0
#>
#> ## Derivatives
#> deriv(S) <- -beta * S * I
#> deriv(I) <- beta * S * I - gamma * I
#> deriv(R) <- gamma * I
```

```r
gen_fn <- function() odin2::odin(code)
sys <- dust_system_create(gen_fn(), list(
  beta = 0.3, gamma = 0.1,
  S0 = 990, I0 = 10, R0 = 0
))
#> -- R CMD INSTALL ------------------------------------------------------------------------------------
#>   -  installing *source* package ‘odin.system935e5e40’ ...
#>    ** this is package ‘odin.system935e5e40’ version ‘0.0.1’
#>      ** using staged installation
#>      ** libs
#>      using C++ compiler: ‘Homebrew clang version 21.1.5’
#>      using SDK: ‘MacOSX15.5.sdk’
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      In file included from dust.cpp:77:
#>    In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/random
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>      /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/rand
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/include/dust2/r/contin
#>       34 |   auto seed = monty::random::r::as_rng_seed<rng_state_type>(r_seed);
#>          |                                         ^
#>    dust.cpp:81:20: note: in instantiation of function template specialization 'dust2::r::dust2_conti
#>       81 |   return dust2::r::dust2_continuous_alloc<odin_system>(r_pars, r_time, r_time_control, r_
#>          |                     ^
#>      2 warnings generated.
```

```
#>      clang++ -arch arm64 -std=gnu++17 -dynamiclib -Wl,-headerpad_max_install_names -undefined dynami
#>      installing to /private/var/folders/yh/30rj513j6mn1n7x556c2v4w80000gn/T/RtmpLWyqXP/devtools_inst
#>      ** checking absolute paths in shared objects and dynamic libraries
#>   -  DONE (odin.system935e5e40) (370ms)
#>
dust_system_set_state_initial(sys)
times <- seq(0, 150, by = 0.5)
res <- dust_system_simulate(sys, times)
state <- dust_unpack_state(sys, res)

df <- data.frame(
  time = rep(times, 3),
  value = c(state$S, state$I, state$R),
  compartment = rep(c("S", "I", "R"), each = length(times))
)

ggplot(df, aes(x = time, y = value, colour = compartment)) +
  geom_line(linewidth = 0.8) +
  labs(title = "SIR via composition (resource sharers)", x = "Time", y = "Count") +
  theme_minimal()
```



5. Stochastic SIR (Petri net)

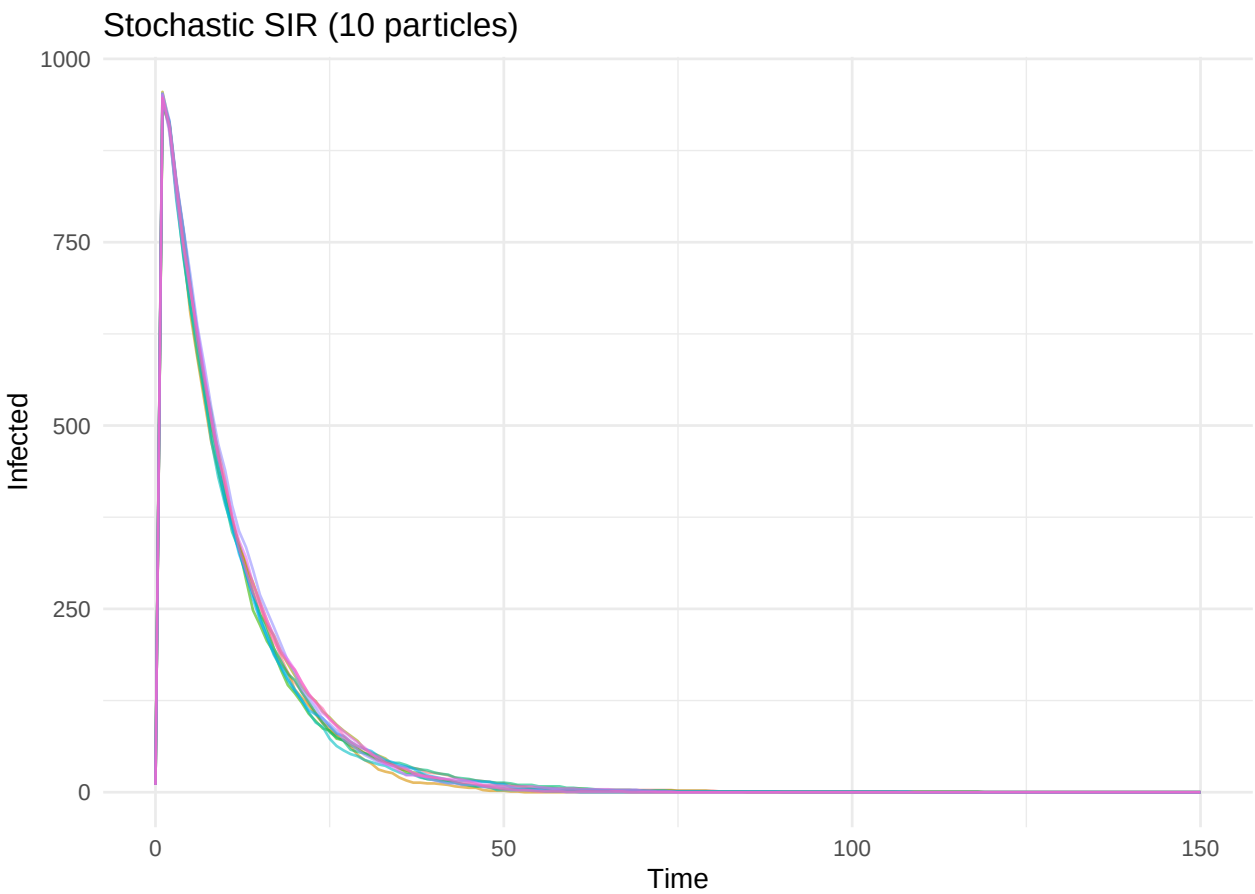
Using odin2's stochastic generation for a discrete-time stochastic SIR:

```r
sir_pn <- labelled_petri_net(
  c("S", "I", "R"),
  infection = c("S", "I") %=>% c("I", "I"),
  recovery = "I" %=>% "R"
)

stoch_code <- pn_to_odin(sir_pn, type = "stochastic")
cat(stoch_code)
#> ## Auto-generated by algebraicodin
#>
#> ## Parameters
#> infection <- parameter()
#> recovery <- parameter()
#>
#> ## Initial conditions
#> S0 <- parameter()
#> I0 <- parameter()
#> R0 <- parameter()
#>
#> initial(S) <- S0
#> initial(I) <- I0
#> initial(R) <- R0
#>
#> ## Transition probabilities and draws
#> p_infection <- 1 - exp(-(infection * I) * dt)
#> n_infection <- Binomial(S, p_infection)
#> p_recovery <- 1 - exp(-(recovery) * dt)
#> n_recovery <- Binomial(I, p_recovery)
#>
#> ## Update equations
#> update(S) <- S - n_infection
#> update(I) <- I + n_infection - n_recovery
#> update(R) <- R + n_recovery
```

```r
gen <- pn_to_odin_system(sir_pn, type = "stochastic")
sys <- dust_system_create(gen(), list(
  S0 = 990, I0 = 10, R0 = 0,
  infection = 0.3, recovery = 0.1, dt = 0.1
), n_particles = 10)
#> -- R CMD INSTALL ------------------------------------------------------------------------------
#>   -  installing *source* package ‘odin.systemc19734ba’ ...
#>    ** this is package ‘odin.systemc19734ba’ version ‘0.0.1’
#>      ** using staged installation
#>      ** libs
#>      using C++ compiler: ‘Homebrew clang version 21.1.5’
#>      using SDK: ‘MacOSX15.5.sdk’
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      In file included from dust.cpp:82:
#>    In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/random
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>      /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/rand
```

```
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/include/dust2/r/discre
#>       41 |   auto seed = monty::random::r::as_rng_seed<rng_state_type>(r_seed);
#>          |                                         ^
#>    dust.cpp:86:20: note: in instantiation of function template specialization 'dust2::r::dust2_discr
#>     86 |   return dust2::r::dust2_discrete_alloc<odin_system>(r_pars, r_time, r_time_control, r_n_pa
#>          |                     ^
#>      2 warnings generated.
#>      clang++ -arch arm64 -std=gnu++17 -dynamiclib -Wl,-headerpad_max_install_names -undefined dynami
#>      installing to /private/var/folders/yh/30rj513j6mn1n7x556c2v4w80000gn/T/RtmpLWyqXP/devtools_inst
#>      ** checking absolute paths in shared objects and dynamic libraries
#>   -  DONE (odin.systemc19734ba) (409ms)
#>
```

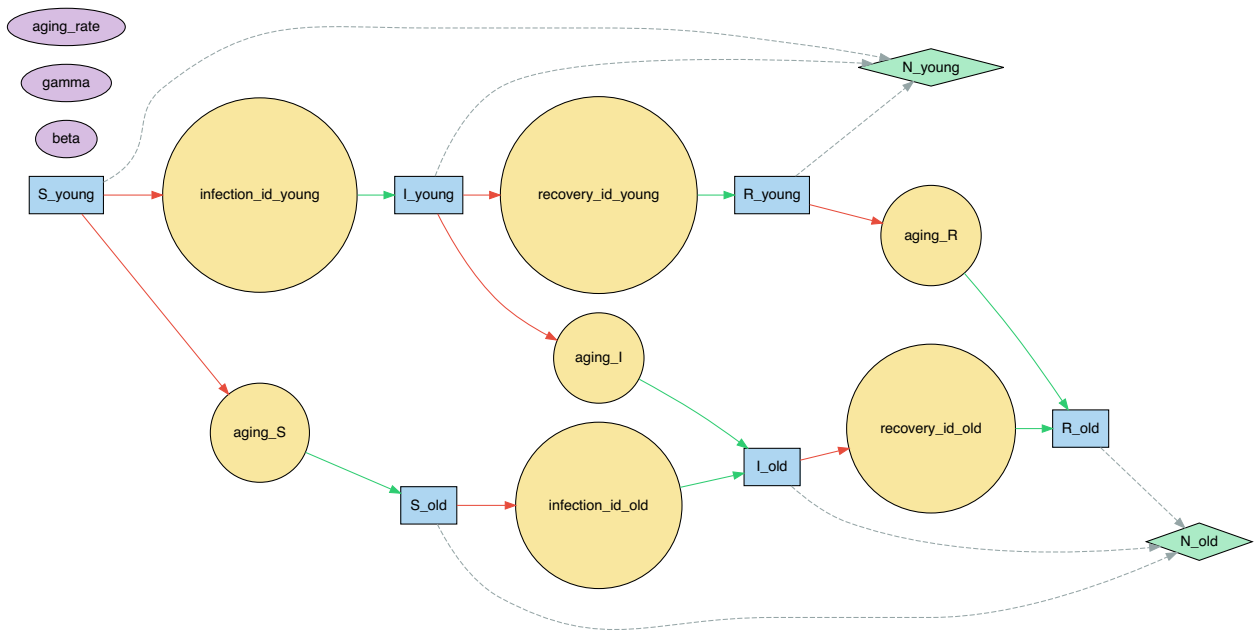
```r
dust_system_set_state_initial(sys)
times <- seq(0, 150, by = 1)
res <- dust_system_simulate(sys, times)
state <- dust_unpack_state(sys, res)

# Plot multiple particles for I
df_list <- lapply(1:10, function(p) {
  data.frame(time = times, I = state$I[p, ], particle = factor(p))
})
df <- do.call(rbind, df_list)

ggplot(df, aes(x = time, y = I, group = particle, colour = particle)) +
  geom_line(alpha = 0.6) +
  labs(title = "Stochastic SIR (10 particles)", x = "Time", y = "Infected") +
  theme_minimal() +
  theme(legend.position = "none")
```



6. Age-stratified SIR (stock-flow stratification)

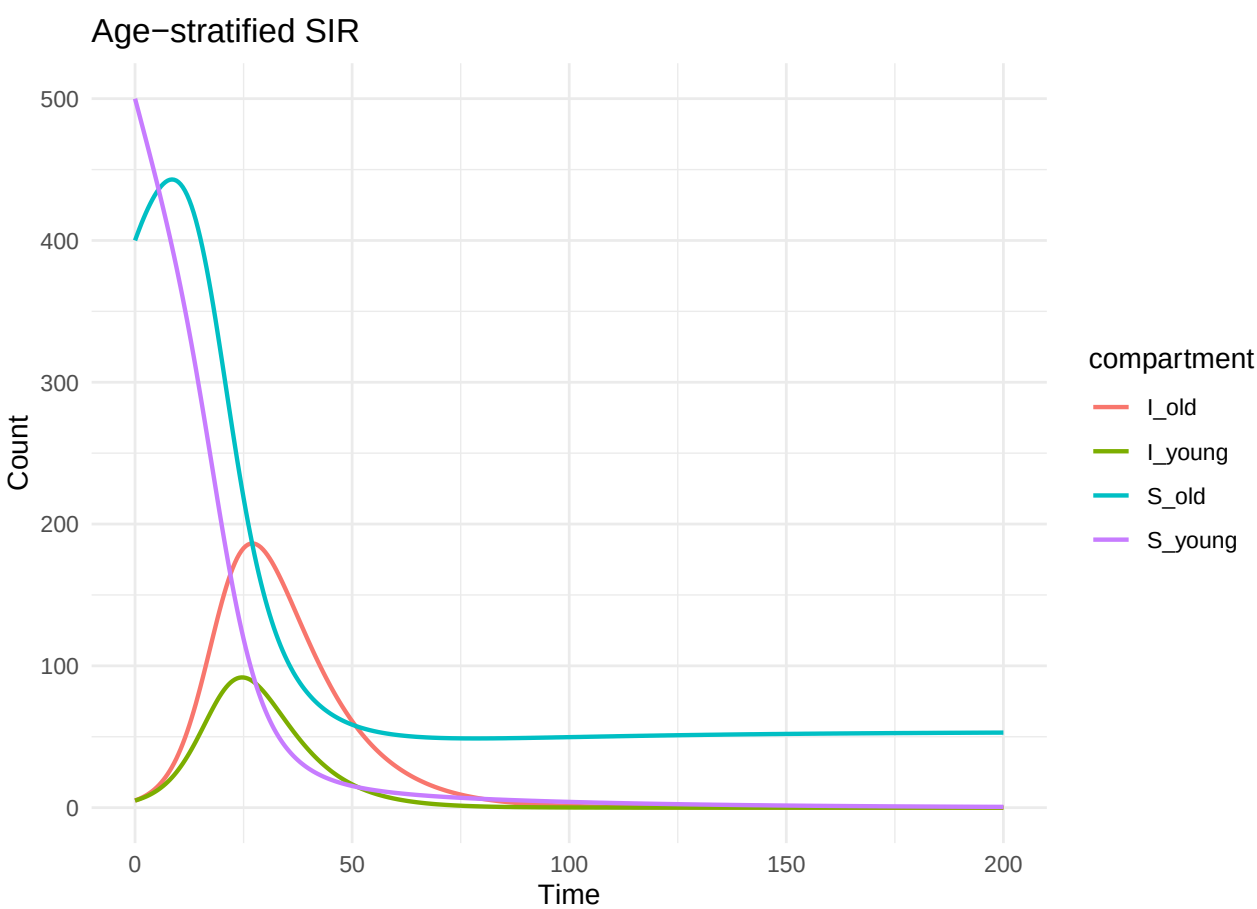
Using the stratification machinery:

```
sir_sf <- stock_and_flow(
  stocks = c("S", "I", "R"),
  flows = list(
    infection = flow(from = "S", to = "I", rate = beta * S * I / N),
    recovery = flow(from = "I", to = "R", rate = gamma * I)
  ),
  sums = list(N = c("S", "I", "R")),
  params = c("beta", "gamma")
)

sir_age <- sf_stratify(sir_sf, c("young", "old"),
  flow_types = c(infection = "disease", recovery = "disease"),
  cross_strata_flows = list(
    list(name = "aging", from = "young", to = "old",
         rate = quote(aging_rate * young), params = c("aging_rate"))
  )
)

plot_stock_flow(sir_age)
```




```
gen <- sf_to_odin_system(sir_age, type = "ode")
sys <- dust_system_create(gen(), list(
  beta = 0.3, gamma = 0.1, aging_rate = 0.02,
  S_young0 = 500, S_old0 = 400,
  I_young0 = 5, I_old0 = 5,
  R_young0 = 0, R_old0 = 0
))
#> -- R CMD INSTALL ------------------------------------------------------------------------------------
#>   -  installing *source* package ‘odin.systemaea97e8a’ ...
#>    ** this is package ‘odin.systemaea97e8a’ version ‘0.0.1’
#>      ** using staged installation
#>      ** libs
#>      using C++ compiler: ‘Homebrew clang version 21.1.5’
#>      using SDK: ‘MacOSX15.5.sdk’
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      In file included from dust.cpp:115:
#>    In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/random
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>      /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/rand
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/include/dust2/r/contin
#>       34 |   auto seed = monty::random::r::as_rng_seed<rng_state_type>(r_seed);
#>          |                                            ^
#>    dust.cpp:119:20: note: in instantiation of function template specialization 'dust2::r::dust2_cont
#>    119 |   return dust2::r::dust2_continuous_alloc<odin_system>(r_pars, r_time, r_time_control, r_n_
#>          |                     ^
#>      2 warnings generated.
#>      clang++ -arch arm64 -std=gnu++17 -dynamiclib -Wl,-headerpad_max_install_names -undefined dynami
#>      installing to /private/var/folders/yh/30rj513j6mn1n7x556c2v4w80000gn/T/RtmpLWyqXP/devtools_inst
#>      ** checking absolute paths in shared objects and dynamic libraries
```

```
#>   -  DONE (odin.systemaea97e8a) (431ms)
#>
dust_system_set_state_initial(sys)
times <- seq(0, 200, by = 0.5)
res <- dust_system_simulate(sys, times)
state <- dust_unpack_state(sys, res)

df <- data.frame(
  time = rep(times, 4),
  value = c(state$I_young, state$I_old, state$S_young, state$S_old),
  compartment = rep(c("I_young", "I_old", "S_young", "S_old"),
                    each = length(times))
)

ggplot(df, aes(x = time, y = value, colour = compartment)) +
  geom_line(linewidth = 0.8) +
  labs(title = "Age-stratified SIR", x = "Time", y = "Count") +
  theme_minimal()
```

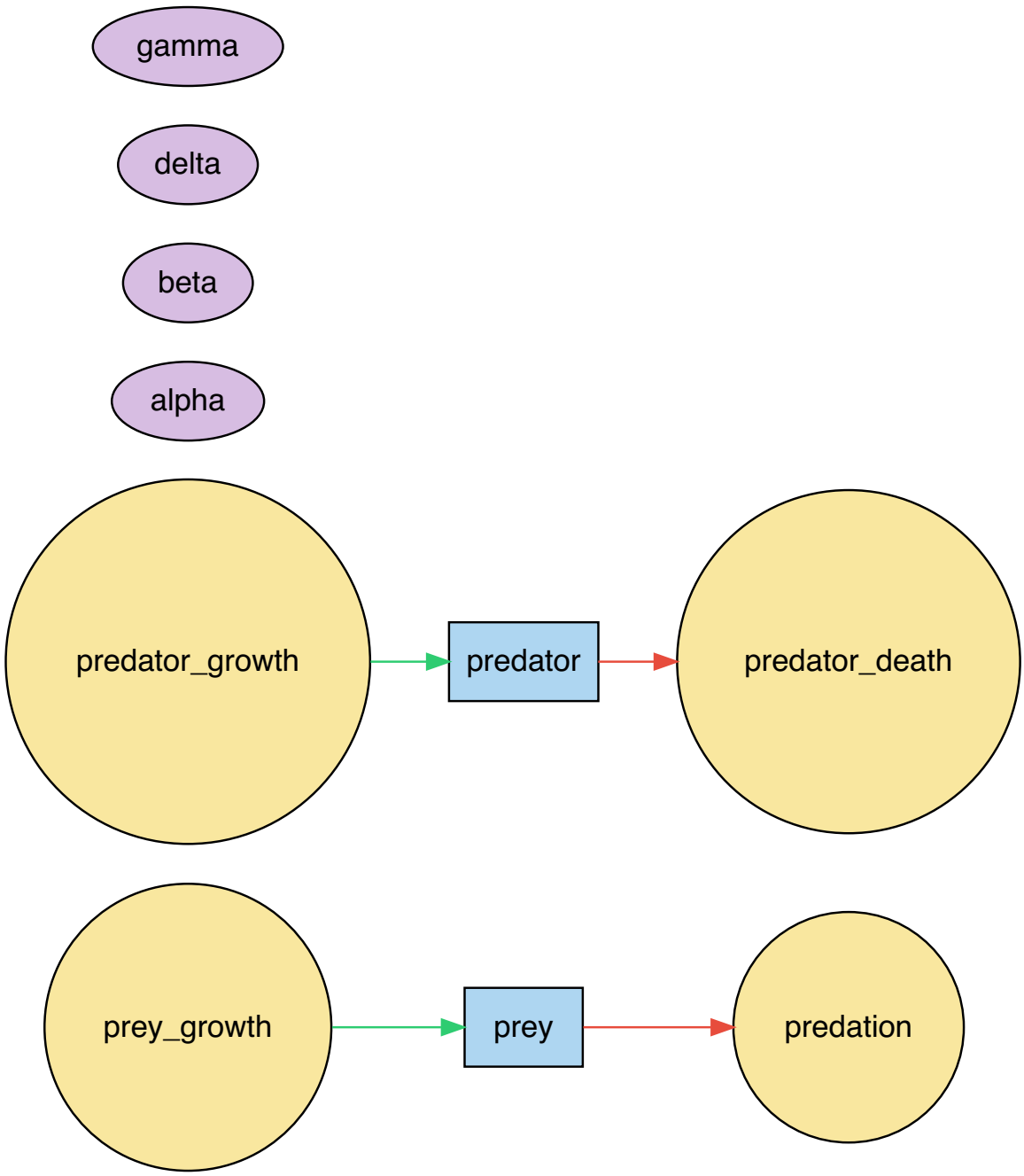


7. Lotka-Volterra predator-prey

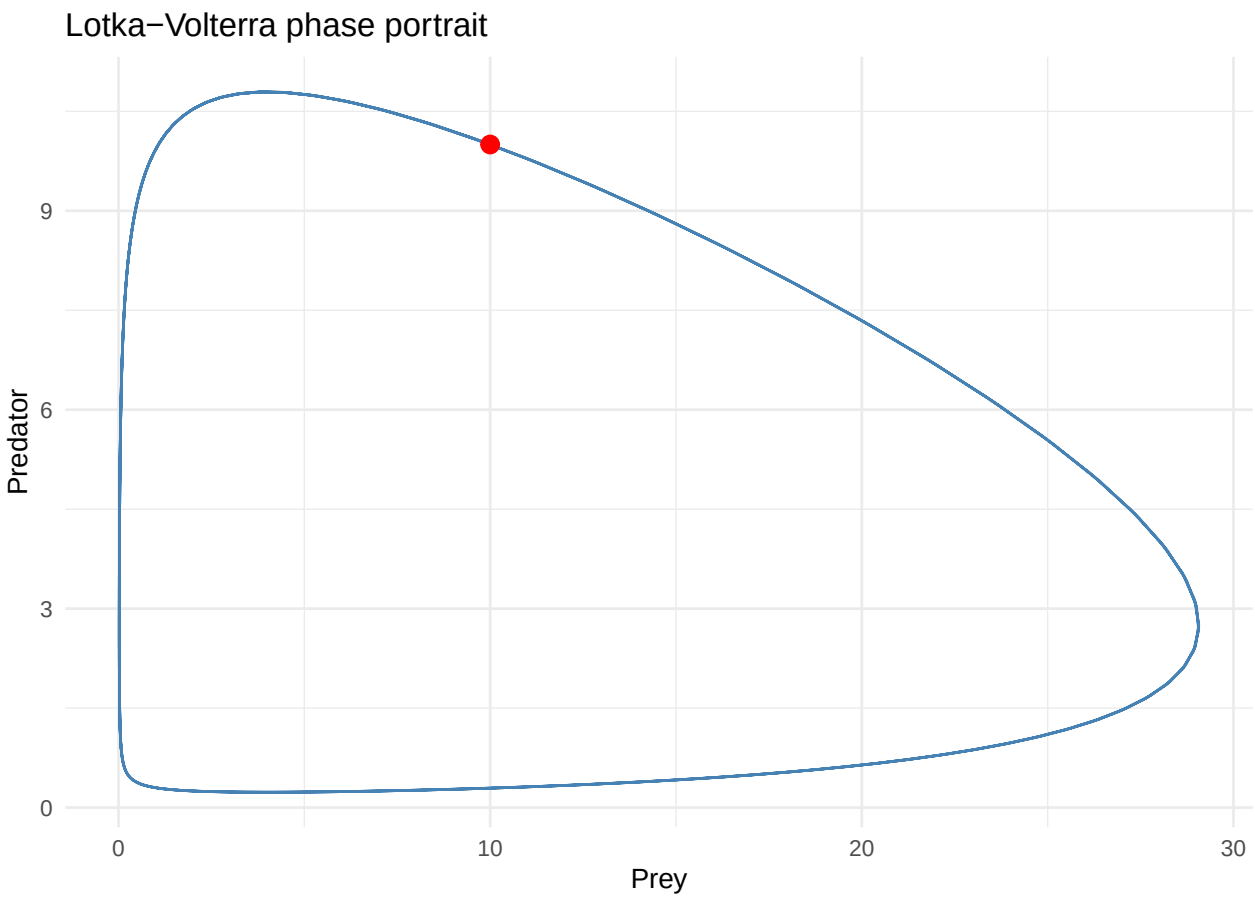
Stock-flow models aren't limited to epidemiology. Here's the classic Lotka-Volterra predator-prey system:

```
lv <- stock_and_flow(
  stocks = c("prey", "predator"),
```

```r
  flows = list(
    prey_growth = flow(from = NULL, to = "prey", rate = alpha * prey),
    predation = flow(from = "prey", to = NULL, rate = beta * prey * predator),
    predator_growth = flow(from = NULL, to = "predator",
                           rate = delta * prey * predator),
    predator_death = flow(from = "predator", to = NULL, rate = gamma * predator)
  ),
  params = c("alpha", "beta", "delta", "gamma")
)

plot_stock_flow(lv)
```



```r
gen <- sf_to_odin_system(lv, type = "ode")
sys <- dust_system_create(gen(), list(
  alpha = 1.1, beta = 0.4, delta = 0.1, gamma = 0.4,
  prey0 = 10, predator0 = 10
))
#> -- R CMD INSTALL ------------------------------------------------------------------------------
#>   -  installing *source* package ‘odin.systemeab00875’ ... (456ms)
#>    ** this is package ‘odin.systemeab00875’ version ‘0.0.1’
#>      ** using staged installation
#>      ** libs
#>      using C++ compiler: ‘Homebrew clang version 21.1.5’
#>      using SDK: ‘MacOSX15.5.sdk’
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      In file included from dust.cpp:82:
#>    In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/random
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>      /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/rand
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/include/dust2/r/contin
#>       34 |   auto seed = monty::random::r::as_rng_seed<rng_state_type>(r_seed);
#>          |                                     ^
#> dust.cpp:86:20     dust.cpp:86:20: note: in instantiation of function template specialization 'dust2
#>       86 |   return dust2::r::dust2_continuous_alloc<odin_system>(r_pars, r_time, r_time_control, r_
#>          |                     ^
#>      2 warnings generated.
#>      clang++ -arch arm64 -std=gnu++17 -dynamiclib -Wl,-headerpad_max_install_names -undefined dynami
#>      installing to /private/var/folders/yh/30rj513j6mn1n7x556c2v4w80000gn/T/RtmpLWyqXP/devtools_inst
#>      ** checking absolute paths in shared objects and dynamic libraries
#>   -  DONE (odin.systemeab00875) (371ms)
#>
dust_system_set_state_initial(sys)
times <- seq(0, 50, by = 0.05)
res <- dust_system_simulate(sys, times)
state <- dust_unpack_state(sys, res)

# Phase plot
df_phase <- data.frame(prey = state$prey, predator = state$predator)
ggplot(df_phase, aes(x = prey, y = predator)) +
  geom_path(colour = "steelblue", linewidth = 0.5) +
  geom_point(data = df_phase[1, ], colour = "red", size = 3) +
  labs(title = "Lotka-Volterra phase portrait", x = "Prey", y = "Predator") +
  theme_minimal()
```



Summary

Model	Approach	Key feature
SIRS	Petri net + stock-flow	Waning immunity
SIR demographics	Stock-flow	Open population
SEIR	Petri net	Latent class
Composed SIR	Resource sharers	Compositional
Stochastic SIR	Petri net	Multiple particles
Age-stratified SIR	Stock-flow stratification	Cross-strata aging
Lotka-Volterra	Stock-flow	Non-epi application

All models are generated as odin2 code, compiled to C++ by dust2, and simulated efficiently.